

Shubham Pathneja

Ex-Robotics Intern @IIT MANDI | Team Lead @SAARTHI |
Robotics and AI TIET'27 | ROS | Computer Vision | Robotics |
Simulations | Control Systems | AI/ML/DL |

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Github: <https://github.com/Spathneja21>

Portfolio: <https://shubhampathneja.vercel.app>

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Skills: Programming Tools - Python, C++, Shell scripting **Tools:** ROS1 Noetic, ROS2 Humble, RViz, Gazebo, OpenCV, MuJoCo, Jetson Edge Devices, Rviz, YOLO, TensorFlow, PyTorch, MATLAB, Git, TensorRT, Secure Shell (SSH), Docker.
Technologies: Machine Learning, Computer Vision, Motion Planning, Path Planning, Simultaneous Localization and Mapping (SLAM), Image Processing, Inverse Kinematics, PID and MPC Tuning. **Soft Skills :** Team Work, Leadership, Project Planning

EXPERIENCE & PROJECTS PPT

Robotics Intern, IIT Mandi (CAIR), Himachal Pradesh ([Certificate](#))

May 2026 – July 2026

- Designed and implemented an end-to-end autonomous navigation and perception pipeline for the **Clearpath Husky A200** mobile robot, targeting real-world **last-mile delivery** scenarios. Integrated Autoware's perception and planning stack with NVIDIA TensorRT-optimized inference, deployed and accelerated on a **Jetson AGX Orin** to enable real-time performance under onboard compute constraints, using CUDA for parallel processing of sensor data.
- Built a **GPS-based localization** and tracking module that interfaced with a host website to stream and visualize the robot's live position, densifying the path to translate GPS way-points into smooth, continuous trajectories improving path navigation when the robot makes an error using the MPC controller.
- Developed an obstacle avoidance algorithm fused with the GPS path-tracking system, allowing the robot to dynamically replan or adjust its trajectory around obstacles while still converging back to the intended GPS path.

PROJECTS

Autonomous Exploration Bot, Robotics Lab, TIET (*Ongoing*) ([Github](#))

July 2026 - Present

- Architected an autonomous exploration and SLAM pipeline using sampling based algorithms (RRT*) to identify and navigate towards unexplored frontiers, enabling the Differential drive **Trossen Robotics LoCoBot WX200** 6DOF to map entirely unknown indoor workspaces without prior environmental knowledge.
- Built real-time occupancy mapping using 2D LiDAR for large-scale spatial structure, fused with RGB-D depth camera data for close-range obstacle avoidance and improved navigation.
- Implemented the full pipeline on ROS1 noetic using the Nav2 stack for costmap management and trajectory execution, with **PID controller** tuning for stable motion planning; used Rviz for real-time visualization during development.

Marketplace Navigation Robot Simulator ([Github](#))

March 2026 - June 2026

- Designed and simulated a differential-drive autonomous mobile robot (AMR) in **MuJoCo**, addressing indoor navigation and product-location assistance for large supermarket environments, with realistic wheel dynamics and collision response modeled through MuJoCo's physics engine.
- Implemented **2D LiDAR-based SLAM** to build and continuously update an occupancy map of the supermarket layout (aisles, shelving, walkways), enabling the robot to localize itself and navigate crowded, aisle-structured indoor spaces.
- Integrated the ROS2 Nav2 stack for global path planning to target product locations and local planning/costmaps for real-time obstacle avoidance around dynamic obstacles (customers, carts, staff); used Rviz for visualization and debugging throughout development.

SAARTHI (Smart AI assistant for Task Handling and Information) (*Ongoing*)

June 2025 - Present

- Engineering an intelligent **AI task-scheduling assistant** for college students juggling between coursework, deadlines, and personal commitments, processing tasks described in natural language via OpenAI-whisper and automatically assigning them to optimal time slots based on each student's flexibility and existing schedule.
- Designed a custom weighted scoring model powered by **Reinforcement Learning** to learn and adapt to individual student behavior over time, combined with a **CP-SAT-based optimization engine** (Google OR-Tools) to solve the underlying scheduling problem.
- My Contribution:** Led the team for the product and implemented the **human behavior tracking reinforcement learning** algorithm along with the voice to command generation functionality.
- Tech stack:** Python, NLP, FastAPI, CP-SAT, Reinforcement Learning, PostgreSQL

SafeSight ([Link](#))

January 2026 – April 2026

- Built a real-time behavioral anomaly detection system that can be deployed on existing CCTV infrastructure, implementing **Faster-RCNN with a ResNet50 backbone** to detect and classify anomalous behavior across 3 object classes, achieving 80.71% detection accuracy in live deployment.
- Led the team and architected the end-to-end pipeline from video ingestion through model inference to alerting/visualization and personally implemented the ResNet50 model, optimizing it for fast, real-time inference on camera feeds.
- Integrated the detection model with a **Flask backend** and a **React.js web frontend**, enabling users to monitor live anomaly detections and flagged events through a browser-based interface.

EDUCATION

Bachelors of Engineering in Robotics and Artificial Intelligence

estimated graduation May 2027

Thapar Institute of Engineering and Technology, Patiala, Punjab

- CGPA** - 8.12 / 10.00
- Societies** - Lead as the Head of Designing in Fine Arts and Photography Society, TIET
Served as a core member of Thapar Venture Club, TIET